FNCE30007 Derivative Securities

Tutorial 9 – Delta Hedging

Question 1

What is the delta of a short position in 1000 European call options on silver futures? Both the options and the futures contract mature in eight months. The current futures price is $8 per ounce, the exercise price of the options is $8, the risk-free interest rate is 12% per annum and the volatility of silver is 18% per annum.

Question 2

In question 1, what initial position in silver futures is necessary for delta hedging? If silver itself is used, what is the initial position? Assume no storage costs for silver.
(this question is similar to the last lecture illustration)

Question 3

Discuss the problems that may be experienced when using delta hedging strategies.

Question 4

You have sold a call option on 50,000 shares. You are worried that the price of the shares may rise between now and expiry. You plan to close out your position before the options expire. Explain how you can hedge against this risk.

Question 5

An Australian financial institution has sold 3 month put options on YEN 100 million, with a strike price of 0.01 (ie YEN 1 = AUD 0.01). The current exchange rate is 0.011, the risk-free interest rate in Japan is 1% p.a., the risk-free interest rate in Australia is 6%. The volatility of the YEN is 10%. Advise them on creating a delta neutral portfolio.